



LECTURE 11: Introducing many body physics problems

Up to here we focused on solving the time dependent Schrödinger equation for a single particle. Let us now move to solve the problem of more than one particle taking into account the nature of the particles, i.e. if they are distinguishable or indistinguishable. This brings about quantum statistics, bosons versus fermions and second quantization procedure.

Single particle versus many-body quantum physics

Distinguishable particle - For example, we will consider a very simple model for a molecule: the ionized hydrogen molecule H_2^+ with two protons with coordinates $\mathbf{R}_1$ and $\mathbf{R}_2$ and one electron with coordinate $\mathbf{r}$. The wavefunction to be found there is

$$\psi(\mathbf{R}_1, \mathbf{R}_2, \mathbf{r}).$$

This is a many-body system, but first, electrons and protons are distinguishable. In addition, the protons are fixed pretty well in their position, not allowing for an overlap of their position, which makes the problem separable as well. The solution involves first the introduction of the relative motion coordinate $\mathbf{R} = \mathbf{R}_2 - \mathbf{R}_1$ and the simplification of the desired wavefunction to $\psi(\mathbf{R}, \mathbf{r})$. A Born-Oppenheimer approximation is then performed, which will allow us to solve first the electronic problem for fixed nuclear separation and then the nuclear motion on the resulting potential energy surface.

Indistinguishable particles - A further level of complexity appears in the Helium atom, where two electrons orbit a (fixed) nucleus placed at the origin. The wavefunction then depends on two electronic position operators,

$$\psi(\mathbf{r}_1, \mathbf{r}_2),$$

and now the particles are indistinguishable fermions which share the space. The wavefunction must be anti-symmetric under the exchange of $\mathbf{r}_1 \xleftrightarrow{\text{exchange}} \mathbf{r}_2$ introducing quantum statistics as a fundamental constraint. In addition the electrons also interact with each other, bringing more complexity into the problem.

In the next lectures devoted to aspects of many body physics, we will progress by

- (i) introducing the symmetrization and antisymmetrization of wavefunctions in the first quantization approach
- (ii) introducing the second quantization procedure with accompanying bosonic versus fermionic operators
- (iii) sketching the solution for a two interacting indistinguishable electrons system (the Helium atom)
- (iv) discussing models for electron transport in solids (Hubbard model) and atom transport in optical lattices (Bose-Hubbard model)
- (v) discussing the occurrence of quasi-particles, such as phonons and excitons in solids
- (vi) discussing models of magnetism in solids such as the Ising and Heisenberg models

Many body systems in the first quantization

To build a many body state vector (and wavefunction) one needs to start with the single particle problem. As usual, for a particle of mass m in an external potential $V(\mathbf{r})$ is described by a Hamiltonian

$$\hat{h} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{r}),$$

which has a set of eigenvalues E_n , and eigenfunctions $\phi_n(\mathbf{r})$ (in the position representation) forming an orthonormal basis. The Schrodinger problem then reduces to finding these energies and eigenfunctions of the Hamiltonian and writing the wavefunction's time evolution starting with some initial state.

What happens now when now a few indistinguishable particles find themselves trapped in the same potential? To answer this, let's start simply with two non-interacting particles and write the two-body Hamiltonian as a sum over the two single particle Hamiltonians

$$\hat{H} = \hat{h}_1 + \hat{h}_2 = \hat{h} \otimes \mathbb{1} + \mathbb{1} \otimes \hat{h}.$$

Notice that we properly specify the problem as a tensor product where the occurrence of a hamiltonian applied to the first particle needs to be multiplied to unity for the second one (assuming no interaction). The stationary Schrödinger equation requires to find possible common wavefunctions of the two particles satisfying

$$\hat{H}\psi(\mathbf{r}_1, \mathbf{r}_2) = E\psi(\mathbf{r}_1, \mathbf{r}_2).$$

Indistinguishability comes now into play, requiring that the probability of any state is conserved under exchange of labels

$$|\psi(\mathbf{r}_1, \mathbf{r}_2)|^2 = |\psi(\mathbf{r}_2, \mathbf{r}_1)|^2$$

This means that the wave function can only change by a phase under exchange:

$$\psi(\mathbf{r}_2, \mathbf{r}_1) = e^{i\alpha}\psi(\mathbf{r}_1, \mathbf{r}_2).$$

Exchanging twice must return the original state, therefore $\psi = e^{i2\alpha}\psi$ and thus $e^{i2\alpha} = 1$, giving $\alpha = n\pi$. Only two distinct possibilities exist:

- (i) $\alpha = 0$: $\psi(\mathbf{r}_2, \mathbf{r}_1) = +\psi(\mathbf{r}_1, \mathbf{r}_2)$ (bosons)
- (ii) $\alpha = \pi$: $\psi(\mathbf{r}_2, \mathbf{r}_1) = -\psi(\mathbf{r}_1, \mathbf{r}_2)$ (fermions).

For fermions, putting the two particles in the same one-particle state gives $\psi(\mathbf{r}, \mathbf{r}) = -\psi(\mathbf{r}, \mathbf{r})$, so $\psi(\mathbf{r}, \mathbf{r}) = 0$, which is the Pauli exclusion principle.

Bosons versus fermions

To fulfill the symmetry conditions, two-particle states built from single-particle orbitals ϕ_a, ϕ_b must be symmetrized (bosons) or antisymmetrized (fermions).

Bosons. The symmetrized spatial wavefunction is

$$\psi_B(\mathbf{r}_1, \mathbf{r}_2) = \begin{cases} \frac{1}{\sqrt{2}} [\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_a(\mathbf{r}_2)\phi_b(\mathbf{r}_1)], & a \neq b, \\ \phi_a(\mathbf{r}_1)\phi_a(\mathbf{r}_2), & a = b, \end{cases}$$

so bosons *can* occupy the same single-particle state.

Fermions. The antisymmetric fermionic state is the Slater determinant

$$\psi_F(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_a(\mathbf{r}_1) & \phi_b(\mathbf{r}_1) \\ \phi_a(\mathbf{r}_2) & \phi_b(\mathbf{r}_2) \end{vmatrix} = \frac{1}{\sqrt{2}} [\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_b(\mathbf{r}_1)\phi_a(\mathbf{r}_2)],$$

which vanishes identically if $\phi_a = \phi_b$, expressing the Pauli exclusion principle.

Spin and statistics: bosons vs. fermions

Particles are grouped according to their intrinsic spin:

Bosons. — integer spin ($s = 0, 1, 2, \dots$). They obey Bose–Einstein statistics and have *symmetric* many-body wavefunctions. Examples: photons ($s = 1$), phonons (quasiparticle bosons), ^4He atoms (composite bosons).

Fermions — half-integer spin ($s = \frac{1}{2}, \frac{3}{2}, \dots$). They obey Fermi–Dirac statistics and have *antisymmetric* many-body wavefunctions, enforcing the Pauli exclusion principle. Examples: electrons, protons, neutrons ($s = \frac{1}{2}$), ^3He atoms (composite fermions).

In relativistic quantum field theory, the spin–statistics theorem links the two: integer-spin fields must be quantized as bosons, half-integer-spin fields as fermions.

Two particles in a harmonic trap – constructing the many-body ground state

To illustrate the role of exchange symmetry in many-body systems, consider two identical particles confined in a one-dimensional harmonic oscillator potential

$$\hat{h} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2.$$

The eigenfunctions $\phi_n(x)$ and eigenvalues $E_n = \hbar\omega(n + \frac{1}{2})$ form a complete single-particle basis. For two non-interacting particles the Hamiltonian is

$$\hat{H} = \hat{h}_1 + \hat{h}_2 = \hat{h} \otimes \mathbb{1} + \mathbb{1} \otimes \hat{h}.$$

Bosonic ground state

Identical bosons require a *symmetric* spatial wavefunction. The lowest-energy configuration is both particles occupying the single-particle ground state ϕ_0 . Thus the two-body ground state reads

$$\psi_0^{(B)}(x_1, x_2) = \phi_0(x_1)\phi_0(x_2).$$

To verify its energy, apply $\hat{H}$:

$$\hat{H} \psi_0^{(B)} = (\hat{h}_1 + \hat{h}_2)\phi_0(x_1)\phi_0(x_2) = (\hat{h}\phi_0)(x_1)\phi_0(x_2) + \phi_0(x_1)(\hat{h}\phi_0)(x_2).$$

Using $\hat{h}\phi_0 = E_0\phi_0$,

$$\hat{H} \psi_0^{(B)} = E_0 \phi_0(x_1)\phi_0(x_2) + E_0 \phi_0(x_1)\phi_0(x_2) = (E_0 + E_0)\psi_0^{(B)}.$$

Hence

$$E_0^{(B)} = E_0 + E_0 = \hbar\omega.$$

Fermionic ground state

For two identical fermions the spatial wavefunction must be *antisymmetric*:

$$\psi(x_2, x_1) = -\psi(x_1, x_2).$$

Since $\phi_0(x_1)\phi_0(x_2)$ is symmetric, two fermions cannot occupy the same orbital. The lowest allowed configuration uses ϕ_0 and ϕ_1 . The antisymmetrized two-particle ground state is

$$\psi_0^{(F)}(x_1, x_2) = \frac{1}{\sqrt{2}} [\phi_0(x_1)\phi_1(x_2) - \phi_1(x_1)\phi_0(x_2)].$$

To confirm the energy, apply $\hat{H}$:

$$\hat{H} \psi_0^{(F)} = \frac{1}{\sqrt{2}} \left[(\hat{h}_1 + \hat{h}_2) \phi_0(x_1) \phi_1(x_2) - (\hat{h}_1 + \hat{h}_2) \phi_1(x_1) \phi_0(x_2) \right].$$

Acting term by term:

$$\begin{aligned} (\hat{h}_1 + \hat{h}_2) \phi_0(x_1) \phi_1(x_2) &= E_0 \phi_0(x_1) \phi_1(x_2) + E_1 \phi_0(x_1) \phi_1(x_2), \\ (\hat{h}_1 + \hat{h}_2) \phi_1(x_1) \phi_0(x_2) &= E_1 \phi_1(x_1) \phi_0(x_2) + E_0 \phi_1(x_1) \phi_0(x_2). \end{aligned}$$

Thus

$$\hat{H} \psi_0^{(F)} = (E_0 + E_1) \psi_0^{(F)}.$$

Therefore,

$$E_0^{(F)} = E_0 + E_1 = \hbar\omega \left(\frac{1}{2} \right) + \hbar\omega \left(\frac{3}{2} \right) = 2\hbar\omega.$$

Conclusion: Bosons: both particles occupy ϕ_0 , giving $E_0^{(B)} = \hbar\omega$. Fermions: exclusion forces occupation of ϕ_0 and ϕ_1 , giving $E_0^{(F)} = 2\hbar\omega$.

Second quantization – the bosonic case

Let us now focus on two bosons, each subject to the single-particle Hamiltonian $\hat{h}$ and use the orthonormal set of *eigenfunctions* of the single-particle Hamiltonian

$$\hat{h} \phi_j(x) = E_j \phi_j(x), \quad \int dx \phi_{j'}^*(x) \phi_j(x) = \delta_{j'j}.$$

We denote an abstract ket for each eigenfunction via

$$\phi_j(x) = \langle x | j \rangle.$$

In this eigenbasis, the Hamiltonian is diagonal,

$$\hat{h} = \sum_j E_j |j\rangle \langle j|.$$

In the symmetrized two-particle basis, the bosonic symmetrization condition written earlier at the level of wavefunctions can be expressed directly at the level of kets:

$$|j, j'\rangle_S = \begin{cases} \frac{1}{\sqrt{2}} (|j\rangle |j'\rangle + |j'\rangle |j\rangle), & j \neq j', \\ |j\rangle |j\rangle, & j = j'. \end{cases}$$

Acting with the two-particle Hamiltonian

$$\hat{H}_0^{(1)} = \hat{h}_1 + \hat{h}_2$$

we obtain

$$\hat{H} |j, j'\rangle_S = (E_j + E_{j'}) |j, j'\rangle_S.$$

Thus the first-quantized two-boson Hamiltonian is diagonal in this basis, with eigenvalues given by sums of the single-particle energies.

Creation and annihilation operators

We now describe the same system using occupation numbers instead of explicit two-particle wavefunctions. We begin by introducing the *vacuum state*

$$|0\rangle \equiv |0_0, 0_1, 0_2, \dots, 0_j \dots\rangle,$$

which means that no bosons occupy any of the single-particle eigenstates $|j\rangle$ of $\hat{h}$.

To create or remove particles in these eigenstates, we introduce *bosonic creation and annihilation operators* $\hat{a}_j^\dagger$ and $\hat{a}_j$ with the commutation relations

$$[\hat{a}_j, \hat{a}_{j'}^\dagger] = \delta_{jj'}, \quad [\hat{a}_j, \hat{a}_{j'}] = [\hat{a}_j^\dagger, \hat{a}_{j'}^\dagger] = 0.$$

The vacuum is annihilated by all $\hat{a}_j$:

$$\hat{a}_j |0\rangle = 0.$$

Acting once with $\hat{a}_j^\dagger$ creates one boson in the single-particle eigenstate j :

$$|1\rangle_j \equiv |0_0, 0_1, \dots, 1_j \dots\rangle = \hat{a}_j^\dagger |0\rangle.$$

Repeated action produces n_j bosons in eigenstate j :

$$|n_j\rangle_j \equiv |0_0, 0_1, \dots, n_j \dots\rangle = \frac{(\hat{a}_j^\dagger)^{n_j}}{\sqrt{n_j!}} |0\rangle.$$

To have N bosonic particles in the system means to place $N = n_0 + n_1 + \dots + n_j$ onto the states of the single particle Hamiltonian. Below is an example for a system with two particles.

Two-boson Fock states

For a system with *two bosons*, one has the option of placing the two bosons either in the same state or in different ones. Here are some possible arrangement

$$|2, 0, 0, \dots\rangle = \frac{(\hat{a}_0^\dagger)^2}{\sqrt{2!}} |0\rangle, \quad |1, 1, 0, \dots\rangle = \hat{a}_0^\dagger \hat{a}_1^\dagger |0\rangle, \quad |0, 2, 0, \dots\rangle = \frac{(\hat{a}_1^\dagger)^2}{\sqrt{2!}} |0\rangle, \quad |1, 0, 1, 0, \dots\rangle = \hat{a}_0^\dagger \hat{a}_2^\dagger |0\rangle.$$

Let us now notice that for any arbitrary bosonic Fock state

$$|n_0, n_1, \dots\rangle = \prod_j \frac{(\hat{a}_j^\dagger)^{n_j}}{\sqrt{n_j!}} |0\rangle.$$

the following action is just a number operator action

$$\hat{a}_j^\dagger \hat{a}_j |n_0, n_1, \dots\rangle = n_j |n_0, n_1, \dots\rangle,$$

pulling out the number of particles in that particular state. Since a state with occupation numbers $\{n_j\}$ (over all possible j 's) has energy $E = \sum_j n_j E_j$, the operator reproducing this on all states is

$$\hat{H} = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j.$$

This is the way of writing the Hamiltonian fully with creation and annihilation operators. The situation here is applicable for Bose-Einstein condensation of boson onto the ground state of an externally imposed potential or for superconductivity.

Examples of systems and phenomena treated with second quantization formalism

Here are some examples for applications of many body quantum physics:

- (i) **Electrons in a solid:** interacting electrons in a crystal lead to conductivity, magnetism, band structure, and superconductivity; impossible to use a full N -body wave function, so creation/annihilation operators are essential.
- (ii) **Ultracold atoms in optical lattices:** atoms in periodic laser potentials realize Bose–Hubbard and Fermi–Hubbard models, showing tunneling, interactions, and superfluid–Mott transitions; naturally formulated in second quantization.
- (iii) **Photons in cavities or waveguides:** photon number is not conserved and states are Fock states; nonlinear optics and cavity QED require field operators $\hat{a}^\dagger, \hat{a}$.
- (iv) **Molecules and chemical bonding:** electrons are fermions and require antisymmetrization; molecular orbitals and correlation effects are built from Slater determinants and second-quantized operators.
- (v) **Nuclear many-body physics:** nucleons in shell models interact strongly; occupation-number representation and Slater determinants are standard tools.
- (vi) **Spin systems and quantum magnetism:** Heisenberg, Ising, and XY models describe interacting spins; mappings to bosons or fermions (Holstein–Primakoff, Jordan–Wigner) rely on second-quantized formalism.
- (vii) **Superconductivity:** Cooper-pair formation and the BCS Hamiltonian are naturally expressed using creation and annihilation operators for correlated electron pairs.
- (viii) **Quantum computers and qubit arrays:** multi-qubit Hilbert spaces are tensor products and often mapped to fermionic or bosonic operators; many-body operator language underlies simulation algorithms.
- (ix) **Cold-atom simulators of lattice gauge theories:** matter and gauge fields on sites and links require operator-valued fields; particle number is dynamic and best handled with second quantization.